

Day 2

Git vs GitHub

Git is a version control system used to track changes in code over time. It helps you save versions, compare changes, create branches, and merge work safely.

GitHub is a cloud-based platform for hosting Git repositories. It adds collaboration features like pull requests, issue tracking, access control, and team management.

Main differences

Topic	Git	GitHub
What it is	Version control tool	Online hosting and collaboration platform for Git repos
Where it runs	Mostly on your local machine	In the cloud through a web interface
Purpose	Track and manage code changes	Share code, review changes, and collaborate
Internet needed	Not required for local work	Needed to access hosted repositories

Basic terminology

- **Repo / Repository:** A project folder tracked by Git, containing your code and its history. A GitHub repo is the online version of that project.
- **Commit:** A saved snapshot of changes in your local Git history. Commits let you record progress step by step.
- **Push:** Sending your local commits to a remote repository like GitHub so others can see them.
- **Pull:** Bringing the latest changes from the remote repository down to your local machine.
- **Branch:** A separate line of development used to work on a feature or fix without affecting the main codebase.

- **Merge**: Combining changes from one branch into another.

Collaboration invitation

A **collaboration invitation** on GitHub is a request sent to someone to give them access to a repository. You can invite a person directly by their GitHub account or email, and they must accept the invitation before they can collaborate.

Simple workflow example

1. Create a repo.
2. Make code changes locally.
3. **Commit** your changes.
4. **Push** them to GitHub.
5. Invite collaborators if needed.
6. Collaborators **pull** the latest version, make changes, and push their work back.

One-line memory trick

- **Git** = version control on your computer.
- **GitHub** = Git online for sharing and teamwork.